

Kisan Mitra AI

Cotton Pest & Disease Knowledge Base

Complete Natural & Organic Remedy Reference Guide

Chapter 01: Bacterial Blight · Chapter 02: Curl Virus (CLCuD)

Chapter 03: Fusarium Wilt · Chapter 04: Healthy Cotton Care

Biology · Symptoms · Diagnosis · 12 Remedies per Disease

Dosage Charts · Case Studies · FAQ · Seasonal Calendars · IPM Guide

RAG Knowledge Base for AI-Powered Agricultural Advisory

Sources: ICAR · KVK Advisories · Traditional Practices · CICR Nagpur

Version

3.0 — Comprehensive Edition (~38 pages)

Scope	Cotton crop (<i>Gossypium hirsutum</i>) — 4 class disease reference
Usage	RAG retrieval knowledge base for Kisan Mitra AI chat advisory system
Coverage	Disease biology, identification, 12 remedies per disease, dosage tables, case studies, FAQ
Sources	ICAR, KVK Advisories, CICR Nagpur, Traditional Practices, Field Research Data
Contact	CICR Nagpur: +91-712-2500247 · ICAR Helpline: 1800-180-1551

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Bacterial Blight

Xanthomonas citri pv. *malvacearum* (Smith 1901, Dye 1978)

1.1 Disease Overview & Economic Impact

Bacterial Blight of cotton, locally known across Gujarat and Maharashtra as Patti Jhad Bimari (leaf-shedding disease), is one of the oldest recorded cotton diseases in India, documented in field surveys as far back as the 1890s. It is caused by the gram-negative rod-shaped bacterium *Xanthomonas citri* pv. *malvacearum*. The disease is polycyclic — completing 3-4 infection cycles within a single growing season — making it explosively destructive during wet periods.

Economic surveys across Gujarat, Maharashtra, Telangana, and Karnataka consistently document yield losses of 20-40% in moderately infected fields and complete crop failure in severely affected fields during high-rainfall years. Beyond yield loss, lint quality is downgraded due to stained fibres. Seed-borne transmission ensures the disease persists to the next season if infected seed is reused.

30-40%	10-15 days	3-4	Rs 3,200 Cr
Avg yield loss	Infection to symptoms	Cycles per season	Est. annual loss India

1.2 Pathogen Biology & Life Cycle

Xanthomonas citri pv. *malvacearum* is a motile, rod-shaped bacterium 0.4-0.7 x 1.2-2.4 um in size. It produces characteristic yellow pigments (xanthomonodins) and secretes xanthan gum — an extracellular polysaccharide that protects it from desiccation and aids colonisation of plant tissue. The pathogen overwinters primarily inside infected cottonseed and in infected crop debris that persists in field soil for 6-12 months.

The infection cycle initiates when bacteria ooze from active lesions during wet conditions and disperse through rain splash, wind-driven rain, irrigation water, insects, and contaminated farm tools. Entry into the plant occurs through natural openings — stomata, hydathodes — and wounds. Once inside intercellular spaces, the bacterium multiplies rapidly and spreads systemically through the vascular system in severe cases, causing the characteristic black arm symptom.

Key Epidemiological Facts

- **Primary inoculum:** Infected seed — survives 2-3 years in stored seed at room temperature
- **Secondary inoculum:** Lesion exudate dispersed by rain splash — can travel 2-3 metres per rain event
- **Incubation period:** 7-14 days depending on temperature and leaf wetness duration
- **Optimal conditions:** 28-32C temperature, >75% relative humidity, frequent light rains
- **Seed transmission rate:** 15-40% of seeds from infected parent plants carry the pathogen
- **Soil survival:** 6-12 months in infected crop debris on soil surface

1.3 Symptom Identification — Stage by Stage

On Leaves (most common and earliest symptom):

- Initially small (2-5 mm), water-soaked, translucent spots — best visible by holding leaf against sunlight in early morning
- Spots rapidly turn angular (restricted by leaf veins) — this angular shape is the key diagnostic feature
- Lesion colour progresses from pale green to yellow to dark brown to black
- In humid conditions: yellowish bacterial ooze visible on lesion surface; dries to a white or yellowish crust
- Multiple lesions coalesce forming large irregular necrotic patches covering 30-50% of leaf area
- Surrounding tissue shows chlorosis (yellowing halo) in early infection stages
- Severely infected leaves curl, dry, and shed prematurely causing characteristic defoliation

On Bolls (most economically damaging):

- Dark, water-soaked, circular to irregular spots appear on boll surface
- Spots enlarge and turn black with age; sunken appearance on boll wall
- Infected bolls fail to open fully or open unevenly, reducing harvestable lint
- Internal lint fibres become brown-stained, reducing grade and market value
- Severely infected bolls rot, mummify on plant, or drop prematurely

On Stems and Branches — Black Arm (systemic phase):

- Dark brown to jet-black elongated streaks on main stem, petioles, and branches
- Affected branches wilt suddenly — historically called 'black arm' symptom
- Vascular tissue turns brown in cross-section — cut stem to confirm
- Severely infected branches become brittle and may break at point of lesion

On Seedlings (most vulnerable stage):

- Cotyledon blackening — dark spots on seed leaves appear within 7-10 days of emergence
- Damping-off — seedlings collapse and die at soil level due to stem rot
- Stunted, weak seedlings with poor stand establishment

Kisan Tip: Use a 10x hand lens for definitive field diagnosis. Bacterial blight lesions have a characteristic water-soaked translucent margin even when the centre has dried — this border distinguishes them from fungal spots which have dry, often concentric ring patterns from the beginning.

1.4 Conducive Conditions & Spread Mechanisms

Factor	Highly Favourable	Moderately Favourable	Unfavourable
Temperature	28-34C	22-28C	Below 18C or above 38C
Relative Humidity	>80%	65-80%	Below 55%
Rainfall	Frequent light showers with wind	Moderate rain, calm	Dry period > 10 days
Soil Drainage	Waterlogged / poorly drained	Moderate drainage	Well-drained
Crop Stage	Seedling to squaring	Flowering	Post-boll maturation
Wind Speed	Strong wind-driven rain	Light wind	Calm conditions
Nitrogen Level	Excess nitrogen applied	Balanced NPK	Low nitrogen

1.5 Natural & Organic Remedies (12 Methods)

Arranged from most to least recommended based on ICAR trial efficacy data:

01

Copper Oxychloride (COC) Spray

The gold standard for bacterial blight management with decades of field validation. Copper ions denature bacterial proteins and disrupt cell membranes on contact. Mix 3 g copper oxychloride (50% WP) per litre of water with 0.5 ml liquid soap as a spreader-sticker. Spray both leaf surfaces thoroughly. Begin at first symptom appearance or prophylactically at 30 DAS in high-risk fields. Repeat every 10-12 days. Do not apply above 38C or within 4 hours of expected rain. ICAR trials: 55-70% disease reduction.

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Bordeaux Mixture

Time-tested copper-lime preparation effective for both bacterial and fungal diseases. Preparation: Dissolve 100 g copper sulphate in 5 litres water in a clay vessel. Separately slake 100 g fresh lime in 5 litres water. SLOWLY pour copper sulphate into lime solution while stirring — NEVER reverse the order. Test with iron nail — if it turns copper-coloured, add more lime. Apply within 2 hours of preparation. Repeat every 14 days. Do not store — prepare fresh each time.

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Neem Oil Emulsion

Cold-pressed neem oil contains azadirachtin, nimbidin, and salanin with bacteriostatic and insecticidal properties. For bacterial blight: 5 ml neem oil + 2 ml liquid soap per litre of warm water. Emulsify by stirring vigorously for 2 minutes until milky white. Spray both leaf surfaces. Apply early morning or after 4 PM to prevent leaf scorch. Repeat every 7 days. Most effective as preventive or at very early infection stage.

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Garlic-Chilli Extract Spray

Allicin (garlic) and capsaicin (chilli) together provide strong antibacterial and anti-feedant action. Crush 100 g garlic + 10 g dried red chilli, soak in 1 litre water for 12 hours, strain through fine cloth. Spray undiluted on infected areas. Prepare fresh — allicin degrades within 24 hours. Can be combined with neem oil for synergistic effect. Apply every 5-7 days during active infection.

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***Pseudomonas fluorescens* Biocontrol Spray**

This beneficial bacterium produces antibiotic compounds — 2,4-diacetylphloroglucinol (2,4-DAPG), phenazines — that suppress *Xanthomonas* populations on leaf surfaces. Mix 10 g *Pseudomonas fluorescens* WP per litre water. Spray in evening — UV radiation degrades the bacterium rapidly. Apply every 15 days as part of integrated management. Available from KVKs and bio-input dealers.

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Turmeric + Lime Spray

Curcumin in turmeric is a proven antimicrobial with demonstrated activity against *Xanthomonas* in laboratory assays. Lime raises leaf surface pH making it inhospitable to bacterial survival. Mix 10 g turmeric powder + 5 g slaked lime per litre water. Stir well. Spray both leaf surfaces every 10 days. Leaves may show temporary light yellowing — this is normal and disappears in 2-3 days. Traditional practice validated across Gujarat farming communities.

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Wood Ash Dusting

Dry wood ash raises leaf and soil surface pH while providing potassium, silica, and calcium — all of which strengthen plant cell walls and reduce bacterial penetration. Dust fine sieved wood ash on leaves and around plant base in early morning when dew provides adhesion. Apply after each rainfall event once leaves dry. Particularly effective for resource-limited farmers as it is freely available from cooking fires. Repeat after every rain.

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Trichoderma viride Foliar Application

Trichoderma viride produces volatile antibacterial compounds (harzianic acid, harzianolide) when applied as foliar spray. It colonises leaf surfaces preventively before pathogens can establish. Mix 5 g Trichoderma viride WP per litre water. Spray in evening. Most effective when applied preventively every 15 days from 21 DAS rather than as a curative treatment after symptoms appear.

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Onion + Ginger Extract

Quercetin and organosulphur compounds in onion combined with gingerols in ginger provide synergistic antibacterial action. Blend 150 g onion + 50 g fresh ginger in 500 ml water. Strain thoroughly. Dilute 1:5 with water. Spray on infected and surrounding healthy leaves every 7 days. Widely practised in Vidarbha and Marathwada regions with positive outcomes.

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Cow Urine (Gomutra) Spray

Fermented cow urine (aged 7-10 days in clay pot) contains urea, uric acid, hippuric acid, phenols, and micronutrients with demonstrated antimicrobial activity. Dilute 1:10 with water and spray on plants. Fresh urine is less effective than fermented. Apply every 10-15 days. Multiple ICAR studies have documented 20-30% disease reduction from cow urine-based sprays. It also acts as a nutritional supplement.

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Hot Water Seed Treatment

Before sowing: submerge seeds in water maintained at exactly 56C for 10 minutes. Use a thermometer — temperature is critical. Below 52C is ineffective; above 58C damages germination. After treatment, dry seeds in shade for 24 hours. This kills 95-100% of seed-borne Xanthomonas without any chemical treatment. This is a preventive measure — the most important single action to reduce bacterial blight incidence.

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Sanitation — Remove, Bag and Burn

At first symptom: prune infected leaves, bolls, and branches at least 5 cm below the last visible lesion. Immediately place pruned material in a bag — do not lay it in the field. Transport to field boundary and burn. Sanitise pruning tools between every plant with 1% bleach solution or 70% rubbing alcohol. This single practice reduces secondary inoculum load by 40-60% and dramatically slows epidemic spread.

1.6 Dosage & Application Schedule

Remedy	Dose per Litre	Interval	Best Time	Both Surfaces?
Copper Oxychloride	3 g (50% WP) + 0.5 ml soap	10-12 days	Morning (dry)	YES
Bordeaux Mixture	1% (100g CuSO ₄ + 100g lime/10L)	14 days	Dry morning	YES
Neem Oil	5 ml + 2 ml soap	7 days	Early morning / Evening	YES
Garlic-Chilli Extract	100 g+10 g per litre (undiluted)	5-7 days	Morning	YES — infected areas
Pseudomonas fl.	10 g WP	15 days	Evening (no UV)	YES — full canopy
Turmeric + Lime	10 g + 5 g	10 days	Morning	YES
Wood Ash	Dust — no dilution	After each rain	Early morning (dew)	Leaves + base
Trichoderma v.	5 g WP	15 days	Evening	YES — full canopy
Onion-Ginger Extract	1:5 dilution	7 days	Morning	YES
Cow Urine	1:10 dilution	10-15 days	Morning	YES

1.7 Prevention & Cultural Practices

- **Certified seed only:** Never reuse seed from an infected crop. Source certified, disease-free seed with valid tag
- **Hot water seed treatment:** 56C for 10 minutes kills seed-borne bacteria — mandatory in high-risk areas
- **Field sanitation post-harvest:** Deep-plough field immediately after harvest to bury infected debris below 20 cm
- **Plant spacing:** 75-90 cm between rows, 45 cm between plants — improves air circulation and reduces leaf wetness
- **Irrigation method:** Use drip or furrow irrigation exclusively — overhead/sprinkler irrigation is the fastest disease spreader
- **Nutrient balance:** Excess nitrogen creates succulent, highly susceptible tissue — maintain balanced NPK; emphasise potassium
- **Wind management:** Plant 2-3 rows of bajra or jowar as wind breaks on field boundary to reduce wind-driven rain spread

- **Field access rules:** Prohibit field entry during and for 2 hours after rainfall — bacteria spread easily by contact
- **Tool hygiene:** Sanitise all cutting tools with 1% bleach between uses during active disease
- **Crop rotation:** Rotate with non-host crops (wheat, chickpea, sorghum) for minimum 2 seasons in heavily infected fields
- **Varietal selection:** LH 900, NHH 44, MCU 12, Pratap show moderate tolerance — consult local KVK for region-specific advice

1.8 Resistance Management

Exclusive reliance on copper-based sprays over many seasons can select for copper-tolerant strains of *Xanthomonas*. To manage this risk and maintain long-term efficacy:

- Rotate copper-based sprays with biological controls (*Pseudomonas*, *Trichoderma*) and organic extracts (neem, garlic)
- Never exceed recommended copper oxychloride dosage (3 g/L) — higher doses accelerate tolerance development
- Integrate cultural practices (sanitation, spacing, rotation) to reduce reliance on any single spray
- Monitor spray efficacy — if standard doses stop working as expected, consult KVK immediately
- Alternate Bordeaux mixture and copper oxychloride in different seasons — do not use both in same season

1.9 Field Case Study — Amreli District, Gujarat (2022)

"We lost nearly half our crop in 2021 to bacterial blight. In 2022 we started copper oxychloride sprays on day 30 — even before symptoms appeared — combined with neem oil every week and removing infected leaves immediately. Disease was controlled by over 85%. Yield improved from 8 quintals to 18 quintals per acre."

A 15-farmer cooperative in Amreli implemented a structured bacterial blight management program under KVK Amreli guidance in 2022. Protocol: (1) hot water seed treatment, (2) preventive copper oxychloride spray at 30 DAS, (3) weekly neem oil spray from 45 DAS, (4) immediate removal of symptomatic plant parts, (5) drip irrigation replacing sprinkler. Average yield increased from 9.2 to 17.8 quintals/acre — a 93% improvement — with zero chemical pesticide input.

1.10 Farmer FAQ

Q: Can bacterial blight spread from one field to a neighbouring field?

A: Yes. Wind-driven rain is the primary long-distance spread mechanism. Bacteria can travel 50-100 metres in heavy rain events. Early treatment in your field benefits both you and your neighbours.

Q: My field shows symptoms in one corner only — should I treat the whole field?

A: Treat the entire field immediately. Bacteria spread rapidly and invisible early infections likely exist across the field. Spot treatment only delays the problem.

Q: Is copper oxychloride safe for the environment?

A: At recommended rates (3 g/L), it is considered safe. Excessive use can accumulate in soil — always use recommended doses and rotate with biological sprays.

Q: How do I confirm bacterial blight vs. other leaf spots?

A: Three confirmation signs: (1) lesions are angular — bounded by veins, (2) water-soaked translucent margin even on dry lesions, (3) bacterial ooze or crust on lesion surface after a wet period. If all three present, it is almost certainly bacterial blight.

Q: Can I use both copper oxychloride and Bordeaux mixture?

A: Use one or the other — not both simultaneously. Both are copper-based. Alternate them with biological controls (*Pseudomonas*, neem oil) rather than stacking two copper products, which accelerates copper tolerance in the pathogen.

Warning: NEVER mix copper oxychloride with alkaline materials (lime, ash) in the same spray tank — this inactivates both. Apply on alternate days.

Curl Virus (CLCuD)

Cotton Leaf Curl Disease — Begomovirus complex, family Geminiviridae

2.1 Disease Overview & History in India

Cotton Leaf Curl Disease (CLCuD) is arguably the most feared cotton disease in South Asia. First reported in Multan, Pakistan in 1967, it became epidemic in the 1990s — causing losses exceeding USD 5 billion. In India, the disease became epidemic in Punjab in the late 1990s and has progressively spread to Haryana, Rajasthan, and Gujarat. CLCuD is caused by a complex of begomoviruses transmitted exclusively by the whitefly *Bemisia tabaci*. There is NO direct remedy that eliminates the virus from an infected plant.

The genetic plasticity of the CLCuD virus complex makes it particularly formidable. New viral strains continuously emerge through mutation and recombination. The New Delhi strain (CLCuNDV) that emerged around 2001 broke the resistance of previously tolerant varieties within 3-4 seasons. The Burewala strain similarly overcame another generation of resistance. This evolutionary arms race means that no currently available variety can be considered permanently resistant. Vector management — controlling *Bemisia tabaci* — is the only effective and durable management strategy.

100%

No cure once infected

30 min

Virus transmission time

\$5B+

Pakistan losses 1990s

600+

Weed host species of vector

2.2 Virus Biology & Whitefly Vector

CLCuD is caused by monopartite single-stranded circular DNA begomoviruses transmitted by *Bemisia tabaci* in a persistent, circulative manner — meaning the virus enters the whitefly's body through feeding, circulates through the gut to the haemolymph, reaches the salivary glands, and is transmitted when the whitefly subsequently feeds on a healthy plant. The virus does NOT replicate inside the whitefly but remains infectious for the insect's entire lifespan.

Bemisia tabaci is a remarkably successful invasive pest. A single female lays 100-200 eggs over her 30-40 day lifespan. Under optimal conditions (32-36C, moderate humidity), populations can increase 10-fold in 3 weeks. The whitefly's cryptic life stages — eggs and nymphs cemented to leaf undersides — make early detection by casual observation very difficult.

Bemisia tabaci Life Cycle

- **Egg stage:** 5-7 days; pale yellow, oval, laid on leaf underside on short stalks
- **Nymph stages (4 instars):** 11-14 days total; flattened, scale-like, immobile from 2nd instar
- **Pupal stage:** 5-6 days; last nymphal instar, slightly raised
- **Adult:** 30-40 days lifespan; 1.0-1.5 mm, white powdery wings, yellow body
- **Generations per season:** 10-12 overlapping generations
- **Dispersal range:** Active flight 100-200 m; passive wind dispersal up to 2-3 km

2.3 Symptom Identification & Differential Diagnosis

Primary Symptoms (Diagnostic):

- **Upward leaf curling** along the midrib — the hallmark symptom; leaves curl upward and inward
- **Enations** — leaf-like or vein-like outgrowths on the **UNDERSIDE** of leaf veins — the **DEFINITIVE** diagnostic feature unique to CLCuD
- Leaf thickening with leathery, stiff texture and dark green colouration
- Vein swelling and darkening (vein thickening) visible on both leaf surfaces
- Mosaic and vein-clearing in early stages in some strains
- Cup-shaped inward rolling of leaf blade in severe infection

Plant-Level Symptoms:

- Severe growth stunting — infected plants 30-60% shorter than healthy neighbours
- Shortened internodes giving characteristic bushy, bunched appearance
- Proliferation of axillary buds leading to excessive branching
- Delayed or absent flowering; flower buds shed prematurely
- Dramatically reduced boll set; formed bolls are small, deformed, and fail to open normally
- Complete economic yield loss in plants infected before 45 DAS

Differential Diagnosis:

Feature	CLCuD	Mite Damage	Water Stress	Herbicide Injury
Enations on leaf underside	YES — definitive	NO	NO	NO
Upward leaf curl	YES	YES (downward more common)	YES (wilting)	YES (cupping)
Vein thickening	YES	NO	NO	YES (sometimes)
Stunting	Severe	Mild-moderate	Mild-severe	Moderate

Recovery with water	NO	NO	YES	Partial
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Kisan Tip: Run your finger along the **UNDERSIDE** of a curled leaf midrib. If you feel bumpy vein-like protrusions (enations), it is almost certainly CLCuD. This single check takes 5 seconds and definitively distinguishes viral curl from other causes.

2.4 Epidemiology & Spread Factors

Risk Factor	Risk Level	Explanation
Whitefly population >5 per leaf	CRITICAL	Immediate management required — virus spread guaranteed
Presence of infected plants nearby	CRITICAL	Every infected plant is an active virus source
Proximity to vegetable crops (tomato, brinjal)	HIGH	Major whitefly and virus reservoir hosts
Presence of weed hosts (Lantana, Parthenium)	HIGH	Year-round reservoir for both whitefly and begomovirus
Late sowing (post June 15)	HIGH	Higher ambient whitefly populations in late kharif
Susceptible variety	HIGH	Most commercial BT hybrids are susceptible
Hot dry weather (32-38C, low humidity)	MODERATE	Accelerates whitefly reproduction cycle
Continuous cotton cropping same field	MODERATE	Ratoon plants are permanent virus and whitefly sources

2.5 Whitefly Vector Management — 12 Natural Methods

Since no infected plant can be cured, ALL management is directed at preventing *Bemisia tabaci* from acquiring and transmitting virus to healthy plants:

01

Yellow Sticky Traps — Mass Trapping

The most consistently recommended physical control. Adult whiteflies are powerfully attracted to yellow wavelengths (530-590 nm). Install traps at crop canopy height — 15 traps per acre minimum. Place before crop emergence to intercept migratory adults. Replace every 14-21 days or when trap surface is covered. Count trapped whiteflies weekly — rising counts indicate intensifying pressure and need for additional management.

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Neem Oil Spray — Anti-feedant and Ovicidal

Azadirachtin in neem oil acts as a feeding deterrent, oviposition suppressant, and insect growth regulator. It does not kill adults on contact but prevents feeding and egg laying, disrupting population growth. 5 ml cold-pressed neem oil + 2 ml soap per litre. Spray LEAF UNDERSIDES — where whiteflies live, feed, and lay eggs. Critical timing: early morning when adults are sedentary. Repeat every 7 days.

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3

Verticillium lecanii — Biological Insecticide

An entomopathogenic fungus that infects and kills whitefly adults, nymphs, and eggs. Conidia (spores) adhere to insect cuticle, germinate, and penetrate through the body wall — the insect dies within 3-7 days with visible white fungal growth. Mix 5-10 g per litre. Spray in late evening — UV radiation kills conidia within hours. Requires >70% relative humidity for optimal efficacy. Most effective during kharif monsoon season.

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Beauveria bassiana — Broad Spectrum Bio-insecticide

Another entomopathogenic fungus with broad efficacy against whitefly adults and nymphs. Works on same principle as *Verticillium lecanii*. Mix 5 g per litre. Apply in evening. Can be alternated with *Verticillium lecanii* to prevent development of resistance in whitefly populations. Particularly effective against nymphal stages on leaf undersides.

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Reflective Silver or Aluminium Mulch

Silver reflective plastic mulch laid between crop rows reflects UV light upward, disorienting whiteflies and confusing their visual landing response. Reduces whitefly landings on plants by 40-70% in trials. Most effective in early crop stages before canopy closes (up to 50 DAS). Also conserves soil moisture and suppresses weeds. Reusable for 2-3 seasons.

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Soap Solution Direct Spray

Dissolve 5 g neem soap or castile soap per litre of warm water. Spray DIRECTLY on whitefly colonies on leaf undersides — soap must make direct contact. The soap dissolves the protective waxy cuticle on nymphs causing dehydration and death within 24-48 hours. Adults must be contacted directly. Effective but requires very thorough application and repeat treatment as new nymphs emerge.

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Weed Host Elimination

Weeds are year-round reservoir hosts for both *Bemisia tabaci* and the begomovirus. Key weed hosts: *Lantana camara*, *Parthenium hysterophorus*, *Ageratum conyzoides*, wild tobacco, and black nightshade. Remove all weeds from field AND a 10 metre border zone before planting. Maintain weed-free border throughout season. Each weed plant removed eliminates thousands of whitefly breeding sites and potential virus sources.

08	Maize or Sorghum Border Crop Barrier Plant 3-4 rows of maize, sorghum, or pearl millet on all four sides of the cotton field 2 weeks before cotton sowing. These tall crops act as physical wind barriers reducing whitefly migration from external sources by 30-50%. Maize is preferred as it also provides habitat for <i>Chrysoperla</i> (green lacewing) — a key whitefly predator. Maize serves as a non-host diverting migrating whiteflies away from cotton.
09	<i>Chrysoperla carnea</i> Release — Predator Augmentation The green lacewing <i>Chrysoperla carnea</i> is one of the most effective predators of whitefly eggs and young nymphs. Each lacewing larva consumes 100-200 whitefly eggs or nymphs during its 2-week larval period. Release purchased <i>Chrysoperla</i> eggs at 50,000 per acre when whitefly populations begin rising. Available from bio-input dealers and ICAR centres. Conserve natural populations by avoiding broad-spectrum sprays.
10	Kaolin Clay Particle Film Food-grade kaolin clay applied as a 3-5% suspension creates a white particle film on plant surfaces that acts as a physical and visual irritant to whiteflies — making plant surfaces unrecognisable as hosts. Kaolin particles also physically abrade soft-bodied insects and interfere with feeding. Completely safe, non-toxic, and readily available. Reapply after rain. Works best combined with sticky traps and neem oil.
11	Trap Cropping with Okra (Bhendi) Okra is a highly preferred host for <i>Bemisia tabaci</i> and can be used as a trap crop. Plant 2-3 rows of okra along field borders 2-3 weeks BEFORE cotton sowing. Whiteflies preferentially colonise okra, concentrating the population away from cotton. Monitor okra rows closely — when whitefly count is high, spray or destroy those rows before adults disperse to cotton. Used successfully in Gujarat and Andhra Pradesh.
12	Infected Plant Immediate Removal Every plant showing confirmed CLCuD symptoms (curl + enations) is an active virus reservoir that viruliferous whiteflies will repeatedly visit. Remove ALL infected plants IMMEDIATELY. Uproot completely including roots. Place directly into a bag. Transport to field edge and burn. Do not leave plants lying in the field. Coordinate with neighbouring farmers — simultaneous removal across multiple fields is 3-4x more effective than individual action.

2.6 Monitoring & Action Threshold Table

Week (DAS)	Monitoring Action	Population Status	Management Response
Pre-sowing	Install yellow traps, scout weeds	—	Remove all weed hosts, lay reflective mulch

Week 2-3 (14-21)	Count adults on 20 plants (3 leaves each)	<1/leaf = safe	Preventive neem oil spray
Week 4-5 (28-35)	Count + inspect for curl symptoms	1-2/leaf = caution	Soap spray + neem oil; check for infected plants
Week 6-8 (42-56)	Twice weekly count; enation check	2-5/leaf = alert	Verticillium/Beauveria spray; intensify neem oil
Week 8-10 (56-70)	Daily scout in high-pressure periods	>5/leaf = critical	Remove infected plants; maximum vector management
Week 10-12 (70-84)	Continue monitoring; assess spread	>10/leaf = emergency	Contact KVK; coordinate with neighbours
Post-harvest	Remove crop debris	—	Deep plough; plan variety and sowing date for next season

2.7 Resistant and Tolerant Varieties

Variety	Region	Tolerance Level	Notes
MRC 6301 (Bunny BT)	Gujarat, Maharashtra	Moderate	Most widely grown in Gujarat
Ankur 651	Vidarbha, Telangana	Moderate	Good yield under moderate whitefly pressure
RCH 134 BT	Punjab, Haryana	Moderate	Suitable for North India cotton belt
HS 6	Rajasthan, Gujarat	Moderate-Good	Traditional variety, more genetically stable
LRA 5166	All regions	Good (non-BT)	Old variety, very stable tolerance
CICR CLCuD tolerant lines	Research stations	High (research)	Contact CICR Nagpur for seed availability
F1886 (JKCH)	Gujarat	Moderate	Popular hybrid with acceptable tolerance

Warning: No currently available variety is IMMUNE to CLCuD. 'Tolerant' means the variety produces some yield even when infected. Whitefly vector management remains critical even with the most tolerant variety.

2.8 Regional Outbreak History in India

Year	Region	Severity	Causal Strain
1997-2002	Punjab, Haryana	Severe epidemic — 30-60% crop loss in affected areas	CLCuKoV (Kokhran strain)
2001-2005	Punjab spread to Rajasthan	New strain broke resistance genes	CLCuNDV (New Delhi strain)
2008-2012	Gujarat emerging hotspots	Moderate — confined to certain districts	CLCuNDV spreading south
2015-2018	Gujarat widespread	Significant — major hybrid varieties affected	Multiple strains
2020-2023	Gujarat, Maharashtra, Telangana	Ongoing moderate-high pressure	Mixed strain complex

2.9 Field Case Study — Bathinda District, Punjab (2021)

"In 2020, 70% of our plants had CLCuD by week 8. In 2021 we installed sticky traps before planting, removed all lantana from field borders, sprayed neem oil from week 2 every week, and immediately uprooted any plant showing curl with enations. Only 8% infection at week 8 — same field, same variety, completely different result."

2.10 Farmer FAQ

Q: Can I save a plant that has started showing curl symptoms?

A: No. Once CLCuD symptoms appear, the virus is systemic throughout the entire plant and cannot be eliminated. The only correct action is immediate removal and destruction.

Q: Whiteflies are visible but no curl symptoms yet — should I act?

A: Yes, immediately. Whiteflies can transmit the virus in just 30 minutes of feeding. The window between 'whiteflies present' and 'virus transmitted to many plants' may be shorter than your next field visit.

Q: How many whiteflies per leaf is the danger threshold?

A: Begin management at 1-2 adults per leaf. Above 5 per leaf is a crisis requiring daily management. Above 10 per leaf, contact your KVK or agriculture department immediately.

Q: My neighbour's field has CLCuD but mine doesn't yet — what should I do?

A: Act immediately as if your field is under maximum threat. Install extra yellow traps on the side facing your neighbour. Intensify neem oil spray frequency to every 5 days. Scout daily. Alert your neighbour to remove infected plants — coordinated action is 3-4x more effective.

Q: Can I plant cotton again next year in the same field after a CLCuD outbreak?

A: Yes, but do not allow any ratoon growth after harvest — ratoon plants carry the virus into next season. Deep plough immediately after harvest, remove all debris, and plan your whitefly management strategy before the next sowing season.

Fusarium Wilt

Fusarium oxysporum f. sp. *vasinfectum* (Atk.) Snyder & Hans., 1945

3.1 Disease Overview & Soil Persistence

Fusarium Wilt is a vascular wilt disease caused by the soil-borne fungus *Fusarium oxysporum* f. sp. *vasinfectum* (FOV). It is among the most intractable cotton diseases because the pathogen produces chlamydospores — thick-walled resting structures that persist in soil for 10-20 years even in the absence of a cotton host. A field infested with Fusarium wilt is effectively infested permanently without sustained long-term soil management.

The disease is most severe in sandy, acidic soils (pH <6.5) and is dramatically worsened by co-infection with root-knot nematode *Meloidogyne incognita*. Nematode feeding wounds on cotton roots provide direct entry points for Fusarium, and nematode-induced physiological stress further compromises plant defence responses. This interaction is called the Fusarium wilt-nematode complex. Fields with both organisms can experience 2-4x greater yield loss than fields with Fusarium alone.

10-20 yrs Soil persistence	25-30C Optimal infection temp	3-4x Worse with nematodes	pH <6.5 High-risk soil
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3.2 Pathogen Biology, Spore Types & Infection Pathway

Fusarium oxysporum produces three morphologically distinct spore types that serve different ecological roles:

- **Microconidia:** Small (5-12 x 2.5-3.5 µm), single-celled, produced in massive numbers in the xylem. Primary colonisation and spread within the vascular system
- **Macroconidia:** Large (25-60 x 3-5 µm), sickle-shaped, multi-septate (3-7 cells). Secondary inoculum produced on plant debris and soil surface
- **Chlamydospores:** Thick-walled (8-12 µm), single or paired, highly resistant to desiccation, heat, and biological degradation. Responsible for 10-20 year soil persistence

Infection begins when soil-borne chlamydospores or microconidia germinate in the root rhizosphere, stimulated by root exudates from germinating cotton seedlings. The germinated spores form infection hyphae that penetrate cotton roots through natural openings, root hair tips, or nematode-created wounds. Inside, the fungus grows through the cortex and into the xylem vessels — where it multiplies explosively, secretes toxins (fusaric acid, lycomarasin), and physically blocks water transport with mycelium and polysaccharide gels, causing the characteristic progressive wilting.

3.3 Symptom Identification & Differential Diagnosis

External Symptoms (progressive):

- **Stage 1:** Sudden midday wilting of leaves — plant appears drought-stressed even when soil is adequately moist
- **Stage 2:** Plant initially recovers by next morning — 'progressive wilt' — but wilts again more severely each subsequent day
- **Stage 3:** Permanent wilting without recovery; yellowing progresses from lower leaves upward
- **Stage 4:** Complete leaf necrosis, defoliation, plant death without boll maturation
- In chronic low-level infection: stunted plants with poor boll set, yellowing, reduced vigour without complete death

Internal Symptoms — Definitive Diagnosis:

- Cut stem at base level — chocolate-brown to dark brown discolouration of xylem wood —
DEFINITIVE CONFIRMATION
- Brown discolouration extends upward through stem; cut at multiple levels to assess spread
- Cross-section shows a clear ring of brown xylem vessels — visible to naked eye
- Root system shows brown-black rotting; lateral and feeder roots most severely affected

Differential Diagnosis:

Feature	Fusarium Wilt	Drought Stress	Verticillium Wilt
Vascular browning	Dark chocolate-brown	None	Light tan/olive
Soil moisture	May be adequate	Dry	May be adequate
Root condition	Rotted, brown	Dry, intact	Mild discolouration
Distribution	Random, individual plants	Uniform across dry area	Often field areas
Recovery overnight	Initially yes, then NO	YES if watered	Partially

Kisan Tip: The brown vascular discolouration confirmed by stem-cutting is the single most reliable Fusarium Wilt diagnostic. No other cotton disease shows identical chocolate-brown xylem discolouration. Carry a pocket knife for field diagnosis — cut every wilting plant at the base as a diagnostic routine.

3.4 The Wilt-Nematode Complex

The interaction between *Fusarium oxysporum* f. sp. *vasinfectum* and root-knot nematode *Meloidogyne incognita* is one of the most studied pathogen complexes in cotton pathology. Nematode infection precedes and facilitates *Fusarium* through three mechanisms: (1) physical wounding of roots creating direct entry sites, (2) physiological suppression of root defence responses at feeding sites, and (3) hormonal changes in roots that favour *Fusarium* colonisation.

Diagnosis of Nematode Co-infection: Dig up affected plants and examine roots. Root-knot nematodes cause distinctive round swellings (galls) on roots ranging from 1-15 mm diameter. If galls are present, both organisms need simultaneous management.

Scenario	Typical Yield Loss	Recommended Additional Action
Fusarium alone (no nematodes)	15-30%	Standard Fusarium management protocol
Nematodes alone (no Fusarium)	10-20%	Nematode management only
Both Fusarium + nematodes	40-70%	Simultaneous management of both — highest priority

3.5 Soil & Climate Risk Factors

- **Soil pH <6.5:** Fusarium oxysporum thrives in acidic soils — liming is the most cost-effective single intervention
- **Sandy or sandy-loam texture:** Poor water retention stresses plants and reduces biological competition against Fusarium
- **Waterlogging:** Anaerobic soil conditions suppress Trichoderma and Pseudomonas (biocontrol agents) while Fusarium tolerates low oxygen
- **Root-knot nematodes:** Even low nematode populations (500 per 500 g soil) multiply Fusarium impact dramatically
- **Continuous cotton:** Each season adds chlamydospores — progressive inoculum build-up without rotation
- **Compacted hardpan:** Restricts root depth and creates anaerobic zones below 15-20 cm
- **Low organic matter (<0.5%):** Depleted soil microbiome — weaker biological suppression of Fusarium
- **Temperature 28-32C + moisture stress:** Combined condition is optimal for infection by FOV

3.6 Natural & Organic Remedies (12 Methods)

01

Trichoderma harzianum Soil Inoculation

The most validated and effective biological control against Fusarium wilt worldwide. *T. harzianum* controls Fusarium through multiple mechanisms: direct hyperparasitism (physically attacks and kills Fusarium hyphae), competition for nutrients and colonisation sites in the rhizosphere, and induction of systemic resistance in the cotton plant (ISR).
 Application: Mix 1 kg *T. harzianum* WP with 50 kg well-composted FYM. Incorporate into sowing furrows at planting. Drench at 5 g per litre around root zone at 30 and 60 DAS.
 Seed treatment: 4 g per kg seed as slurry. ICAR trials show 50-70% reduction in wilt incidence.

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2

Pseudomonas fluorescens Rhizosphere Drench

P. fluorescens produces 2,4-diacetylphloroglucinol (2,4-DAPG), phenazine-1-carboxylic acid, and pyoluteorin — antibiotic compounds that strongly suppress *Fusarium* oxysporum germination and growth in soil. Also produces siderophores that deprive *Fusarium* of iron. Drench 250 ml of 10 g/L suspension per plant at the root zone. Apply at planting, 30 DAS, and 60 DAS. Works synergistically with *Trichoderma* — can be combined in same drench application.

0
3

Agricultural Lime Soil Amendment

Fusarium chlamydospore germination is dramatically reduced above pH 6.5. Raising pH from 5.5 to 6.8 can reduce *Fusarium* propagule germination by 60-80%. Get a soil pH test before applying. Typical application rate: 500 kg (pH 6.0-6.5 soils) to 2 tonnes/ha (pH 5.0-5.5 soils). Incorporate into top 20 cm soil 4-6 weeks before sowing. Retest pH each season — avoid overliming above pH 7.5 which causes micronutrient deficiencies.

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4

Neem Cake Soil Incorporation

Neem cake contains nimbidin, nimbidiol, nimbolide, and other antifungal and nematicidal compounds. It addresses BOTH components of the wilt-nematode complex simultaneously. Apply 250-500 kg per hectare incorporated into top 15 cm soil 2-3 weeks before sowing. Compounds are released gradually as the cake breaks down. Also improves soil organic matter and provides balanced slow-release nutrition. Available from oil mills at low cost.

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5

Deep Summer Ploughing + Soil Solarisation

Deep ploughing (30 cm depth) in May-June exposes chlamydospores to lethal UV radiation and desiccation (soil temperatures >45°C at surface). Enhanced solarisation: after deep ploughing, thoroughly irrigate the field, then lay transparent polythene mulch (25-50 micron) on wet soil surface and seal edges. Leave for 4-6 weeks during peak summer. Under the mulch, soil temperature reaches 60-70°C — killing chlamydospores, nematodes, and weed seeds. Most effective long-term management practice.

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6

Mustard Cake Biofumigation Drench

Mustard cake contains glucosinolates that hydrolyse in moist, warm soil to release isothiocyanates — volatile biofumigants with broad-spectrum antifungal activity. Mix 500 g mustard cake per litre water, ferment 48 hours, strain, dilute 1:10. Drench 500 ml per plant around root zone. Most effective when soil is moist and temperature >25°C. Apply 2-3 weeks before sowing for soil treatment or around infected plants during the season.

07

Garlic + Ginger Rhizosphere Drench

Allicin from garlic and gingerols from ginger have demonstrated antifungal activity against *Fusarium oxysporum* in multiple laboratory and greenhouse studies. Blend 200 g garlic + 100 g fresh ginger in 2 litres water. Strain thoroughly through cloth. Dilute 1:10. Drench 500 ml per plant at root zone. Apply every 15 days in affected areas. Cost-effective, readily available input that smallholder farmers can prepare on-farm.

08

Vermicompost Soil Application

Vermicompost contains 10^8 - 10^9 CFU of beneficial bacteria per gram — *Bacillus*, *Pseudomonas*, *Streptomyces* species that are natural *Fusarium* competitors. It also contains humic acids that enhance plant immune responses. IARI studies found vermicompost from *Eisenia fetida* significantly suppresses *Fusarium* in cotton soil. Apply 2 tonnes per hectare as basal dose at sowing.

09

Biochar Soil Amendment

Biochar creates micro-habitats that preferentially support biocontrol organisms like *Trichoderma* and *Pseudomonas* while suppressing *Fusarium* populations. Applied at 2-5 tonnes/ha, biochar improves soil aeration, water-holding capacity, and CEC. Its alkalising effect (biochar pH 8-10) also raises soil pH toward *Fusarium*-suppressive range. Prepare from rice husk, coconut shell, or wood — material must be completely pyrolysed before application.

10

***Paecilomyces lilacinus* — Nematode Biocontrol**

An entomopathogenic fungus that targets root-knot nematode eggs and juveniles in soil. By controlling the nematode component of the wilt-nematode complex, *P. lilacinus* indirectly but significantly reduces *Fusarium* wilt severity. Mix 5 g WP per litre water as soil drench, or incorporate 25 g per kg seed furrow soil at sowing. Can be combined with *Trichoderma harzianum* in the same drench.

11

Cow Urine + Wood Ash Soil Drench

Fermented cow urine (1:10 dilution) combined with leached wood ash tea (soak 500 g ash in 10 L water overnight, strain) creates an alkaline drench with antifungal properties and soil pH adjustment benefit. Apply 1 litre per plant every 15 days. The combination raises localised root-zone pH while providing antimicrobial compounds from urine phenols and potassium from ash.

12

Remove and Deep Bury or Burn Wilted Plants

Non-chemical but critically important. Wilted, dying plants continue to release millions of chlamydospores into surrounding soil as they decay. Remove the entire plant including the full root system — use a spade to extract roots completely. Either burn the plant away from the field or deep-bury at >50 cm depth. After removal, drench the planting hole with *Trichoderma* (10 g/L) and sprinkle lime to prevent the vacated spot from becoming an inoculum hotspot.

3.7 Biological Control Agent Guide

BCA Agent	Dose per Litre	Application Method	Timing	Efficacy Range
Trichoderma harzianum	5-10 g WP	Soil drench + FYM mix + seed coat	Sowing + 30 + 60 DAS	50-70% reduction
Pseudomonas fluorescens	10 g WP	Soil drench + foliar (evening)	Sowing + 30 + 60 DAS	40-60% reduction
Bacillus subtilis	5 g WP	Soil drench + seed coat	Sowing + 30 DAS	30-50% reduction
Paecilomyces lilacinus	5 g WP	Soil drench / furrow application	Sowing + 30 DAS	Targets nematodes — indirect
Glomus intraradices (AM Fungi)	10 g per hole	Root zone at transplanting	At sowing	Improves root health + nutrient uptake
Streptomyces lydicus	5 g WP	Soil drench	Sowing + 30 DAS	Emerging — 30-40% in trials

3.8 Multi-Year Soil Rehabilitation Protocol

For fields with severe, chronic Fusarium wilt — implement this 3-year programme:

Year	Action	Expected Outcome
Year 1 (Off cotton)	Deep plough (30cm), solarise (6 weeks), apply lime, biochar (3 t/ha), sow dhaincha green manure and incorporate	Chlamydospore reduction 40-70%; soil structure improvement; pH correction
Year 1 crop	Grow wheat or chickpea (non-host rotation)	No new Fusarium inoculum added; soil microbiome rebuilds
Year 2 (Off cotton)	Vermicompost (2 t/ha) + Trichoderma + Pseudomonas in FYM; neem cake (300 kg/ha)	Biocontrol establishment; continued inoculum reduction
Year 2 crop	Grow sorghum or maize (non-host rotation)	Continued soil rehabilitation
Year 3 cotton	Resistant variety; full seed treatment protocol; Trichoderma + Pseudomonas drench; monitor weekly	Wilt incidence expected to be <5-10% vs 30-50% before programme

3.9 Field Case Study — Akola District, Maharashtra (2023)

"Our field had Fusarium wilt for 6 years — 20-30% of plants dying every season. We did deep summer ploughing in June, applied 1.5 tonnes lime to raise pH from 5.8 to 6.7, mixed Trichoderma with neem cake and FYM before sowing, and drenched with Pseudomonas at 30 days. This year only 3% plants wilted. It took preparation — but results were clear."

3.10 Farmer FAQ

Q: My field has had wilt for 6 years — can I ever grow cotton again without massive losses?

A: Yes, but a multi-year soil rehabilitation programme (see Section 3.8) is required. With consistent application of lime, deep ploughing, biological control agents, and crop rotation, wilt incidence can be reduced from 30-50% to under 10% within 2-3 seasons.

Q: How do I know if nematodes are also present in my wilt-affected field?

A: Dig up affected plants including complete root system and inspect carefully. Root-knot nematodes create distinctive round swellings (galls) on roots. If galls are visible, you have the wilt-nematode complex and must treat for both simultaneously.

Q: Is there a quick home test I can do to confirm Fusarium in soil?

A: No reliable home test exists. For field confirmation, cut the stem of any wilting plant at the base — chocolate-brown vascular discolouration is effectively diagnostic. For soil confirmation, send 500 g soil to nearest agricultural university or ICAR lab.

Q: Can I replant in a spot where a wilted plant was removed?

A: Only after treatment. Drench the vacated hole thoroughly with *Trichoderma harzianum* (10 g per litre, 2 litres per hole), add a handful of lime, and leave for 7 days before replanting.

Q: Does Trichoderma work if soil is waterlogged?

A: No. *Trichoderma* requires aerobic (oxygenated) soil conditions to survive and colonise roots. Waterlogged, anaerobic soil kills *Trichoderma* while *Fusarium* tolerates low oxygen. Ensure field drainage before biological control applications.

Healthy Cotton Care

Comprehensive agronomic guide for a disease-free, high-yield cotton season

4.1 Indicators of Optimal Plant Health

Understanding what a genuinely healthy plant looks like enables early detection of any deviation toward disease. Most farmers learn to recognise disease — but recognising and actively maintaining health is equally important:

Plant Parameter	Healthy Indicator	Concern — Investigate
Leaf colour	Deep, uniform green throughout canopy	Yellowing, pale, mottled, dark spots
Leaf texture	Smooth, flat, supple, with normal surface sheen	Leathery, curled, thickened, rough
Stem	Firm, well-spaced nodes (8-12 cm)	Short internodes, streaked, hollow, soft
Root (sample 5 plants)	White to cream, well-branched, firm, extending >20 cm	Brown, rotted, galled, sparse lateral roots
Growth rate	8-12 cm per week in active vegetative stage	<5 cm per week without drought cause
Squaring	First square at 40-50 DAS as per variety	Delayed beyond 60 DAS, or absent
Boll retention	>60% of set bolls retained to maturity	Excessive shedding without pest cause
Canopy closure	Rows close by 55-65 DAS	Open, sparse canopy — slow growth

4.2 Soil Health & Pre-Sowing Preparation

Soil health is the ultimate foundation of crop health. A biologically active, well-structured soil with balanced nutrition naturally suppresses pathogens, grows disease-resistant plants, and produces higher yields at lower input cost:

- **Mandatory soil test:** Test pH, NPK, micronutrients (Zn, Fe, Mn, B), organic carbon. Test every 2 years minimum
- **Target pH:** 6.5-7.5 for cotton. Adjust with agricultural lime (too acidic) or elemental sulphur (too alkaline)

- **Organic carbon target:** Minimum 1.0-1.5% OC — below this, biological suppression of pathogens collapses
- **Deep ploughing:** 30 cm depth during summer (May-June) to break hardpan, expose pathogens to UV, improve drainage
- **Green manure:** Sow dhaincha (*Sesbania bispinosa*) or cowpea 45 days before cotton; incorporate at flowering — adds 80-120 kg N/ha equivalent and doubles soil microbial activity
- **FYM application:** 10-15 tonnes well-composted FYM per hectare at least 3 weeks before sowing
- **Bio-inoculant enrichment:** Add *Trichoderma harzianum* + *Pseudomonas fluorescens* + *Azotobacter* to FYM heap 15 days before field application
- **Drainage:** In flat fields, create 30 cm wide drainage channels every 5-6 rows. In low-lying fields, consider raised bed planting
- **Micronutrient correction:** Apply 25 kg ZnSO₄ and 500 g borax per hectare basal — cotton is commonly deficient in both

4.3 Seed Treatment Protocols

Proper seed treatment is the most cost-effective investment per acre — protecting the critical germination and early establishment phase:

Treatment	Method	Dose	Target Protection	Timing
Hot water treatment	56C water, exact temperature	10 minutes	Kill seed-borne bacteria & fungi	24 hrs pre-sowing
<i>Trichoderma viride</i>	Slurry coat, shade dry	4-5 g per kg seed	Soil-borne fungal diseases	Just pre-sowing
<i>Pseudomonas fluorescens</i>	Slurry coat, shade dry	10 g per kg seed	Bacterial & rhizosphere biocontrol	Just pre-sowing
<i>Azotobacter chroococcum</i>	Slurry coat	5 g per kg seed	Nitrogen fixation support	Just pre-sowing
Neem oil soak	Soak in 1% solution	2 hours	Antifungal + pest deterrence	Night pre-sowing
Zinc sulphate	Dry seed coating	2 g per kg seed	Zinc nutrition for germination	At sowing
Phosphate solubilising bacteria	Slurry coat	5 g per kg seed	Phosphorus availability	Just pre-sowing

Kisan Tip: Apply bio-inoculants AFTER neem oil treatment — neem oil can reduce inoculant viability. Apply neem soak on Day 1, dry in shade, apply bio-inoculants on Day 2, sow on Day 2.

4.4 Full-Season Preventive Spray Calendar

Prevention is 5-10x more cost-effective than curative treatment. This calendar protects against all three major cotton diseases simultaneously:

DAS	Spray	Dose per Litre	Target	Notes
21	Neem oil (preventive)	3 ml + 1.5 ml soap	Whitefly, early fungal	Even if no pest visible — prophylactic
28	Trichoderma viride foliar	5 g WP	Fungal disease prevention	Evening spray
30	Copper oxychloride	3 g	Bacterial blight prevention	Prophylactic application
35	Pseudomonas fl. foliar	10 g WP	Bacterial suppression	Evening spray — no UV
42	Neem oil spray	5 ml + 2 ml soap	Whitefly + antifungal	Focus on leaf undersides
49	Garlic-chilli extract	100 g garlic + 10 g chilli per litre	Broad spectrum antimicrobial	Fresh preparation only
56	Copper oxychloride	3 g	Bacterial blight protection	Second prophylactic
60	Trichoderma viride foliar	5 g WP	Fungal prevention	Evening spray
63	Neem oil spray	5 ml + 2 ml soap	Whitefly — critical at squaring	Twice weekly if >2 WF per leaf
70	Turmeric + lime spray	10 g + 5 g	Antimicrobial	Morning spray
80	Bordeaux mixture 1%	100 g CuSO ₄ + 100 g lime per 10 L	Fungal + bacterial protection	Pre-boll formation
90+	Scout-based application	As per diagnosis	Respond to field conditions	Threshold-based only

4.5 Organic Nutrition & Biostimulant Guide

Nutritional stress is a disease gateway. Key nutrients for disease resistance:

Nutrient	Organic Source	Rate & Timing	Disease Resistance Role
Potassium (K)	Wood ash, SOP, granite dust	80-100 kg K ₂ O/ha basal	Most critical — strengthens cell walls, lignification, pathogen resistance
Calcium (Ca)	Agricultural lime, gypsum	2-3 tonnes lime/ha basal (if pH correction needed)	Cell wall integrity, reduced bacterial penetration
Silicon (Si)	Rice hull ash, wollastonite	2 g/L foliar spray at 30, 60 DAS	Hardens leaf cuticle — proven mechanical barrier to pathogens
Zinc (Zn)	ZnSO ₄ , neem-chelated zinc	25 kg ZnSO ₄ /ha basal; 2 g/L foliar at 30 DAS	Enzyme cofactor for plant immune response
Boron (B)	Borax (Na ₂ B ₄ O ₇)	500 g borax/ha basal; 0.5 g/L foliar at flowering	Cell wall formation; pollen viability
Jeevamrit	Cow dung 10 kg + urine 5 L + jaggery + besan in 200 L water, fermented 48 hrs	200 L/ha soil drench at 30, 60, 90 DAS	Massively boosts soil microbiome — 10 ⁹ bacteria/ml
Panchagavya	Cow dung, urine, milk, curd, ghee fermented 15 days	3% foliar spray every 15 days	Immune stimulant, micronutrient source, microbiome support
Seaweed extract	Commercial liquid seaweed concentrate	3 ml/L foliar at squaring and early boll	Cytokinin + auxin content boosts cell division; systemic resistance

4.6 Integrated Pest Management (IPM)

- **Monitoring-based approach:** Scout before spraying — never spray on a calendar alone. Observe, count, then decide
- **Economic Threshold Levels (ETL) — Act only when exceeded:**
 - Whitefly: 2 adults per leaf
 - Aphid: 25% of leaves showing active colony
 - Thrips: 10 thrips per leaf during seedling; 20 at squaring
 - Bollworm: 8-10% squares or bolls damaged
 - Jassid (leafhopper): 2 per leaf during vegetative; 4 at boll stage
- **Biological control first:** Deploy Chrysoperla, Trichogramma, Beauveria, Verticillium before any other spray
- **Trap crops:** Okra planted 3 weeks before cotton diverts bollworm and whitefly away from field
- **Light traps:** 1 trap per 2 acres — attracts and kills nocturnal moths (Helicoverpa, pink bollworm)

- **Pheromone traps:** Species-specific lures for *Helicoverpa armigera*, pink bollworm at 4-5 per acre
- **Flowering plants:** Coriander, fennel, marigold, cowpea rows within field — habitat for natural enemies
- **Record all interventions:** Date, spray used, pest count before and after — builds decision-making knowledge over seasons

4.7 Full Season Agronomic Calendar

Period	DAS	Critical Actions	Disease Watch
Pre-Sowing	-45 to -15	Soil test, deep plough, FYM + bio-inoculants, lime if pH <6.5, green manure incorporation	Soil pathogen assessment
Sowing	0	Full seed treatment protocol, correct spacing (75x45 cm), adequate pre-sow moisture	—
Germination	1-10	Monitor germination %; gap-fill by day 10; check for damping-off	Damping off
Early Seedling	11-20	Thinning to 1 plant per hill, first scouting for pests, install sticky traps	Seedling blight
Late Seedling	21-35	First neem oil spray (21 DAS), weed control, first Trichoderma foliar	Early curl virus signs
Early Vegetative	36-50	Copper oxychloride (30 DAS), Pseudomonas spray (35 DAS), neem oil (42 DAS)	Bacterial blight, wilt
Squaring	51-70	HIGHEST RISK PERIOD — intensify all monitoring. Neem oil twice weekly if WF>2 per leaf	All diseases at peak
Flowering	71-85	Bordeaux mixture (80 DAS), monitor boll set, manage whitefly aggressively	CLCuD, boll rot

Early Boll	86-110	Scout for boll rot, maintain potassium nutrition, reduce overhead irrigation	Boll rot, late wilt
Boll Maturation	111-130	Reduce irrigation, allow natural defoliation, monitor for cotton stainer bug	Fibre quality issues
Harvest	130+	Harvest promptly when 60% bolls open — delayed harvest reduces lint quality	Staining, weathering
Post-Harvest	—	Deep plough immediately, remove all debris, plan rotation and cover crop	Disease carryover

4.8 Scouting Protocol & Record Keeping

Weekly Scouting Protocol (15-20 minutes per acre):

- **Scouting pattern:** Walk field diagonally in a W-pattern. Stop at 5 random points. Inspect 4 plants per stop = 20 plants total
- **At each plant:** Check 3 leaves — upper, middle, lower canopy. Count whitefly adults on underside of middle leaf
- **Check for:** Leaf spots (bacterial blight), leaf curl + enations (CLCuD), wilting + stem browning (Fusarium), bollworm damage
- **Dig check:** Once every 2 weeks, gently extract 2 plants to examine roots for rot, galls, discolouration
- **Record in notebook:** Date, growth stage, pest counts, disease % if any, weather, interventions planned

Annual Records Every Farmer Should Maintain:

- Sowing date, variety name, seed source, seed lot number, seed treatment applied
- All inputs applied — organic matter, bio-inoculants, sprays — with exact dates and rates
- Disease incidence records: % plants affected per disease, at peak and end of season
- Pest population counts from scouting — builds understanding of seasonal patterns
- Yield per acre (quintals) and lint grade and price received
- Soil test results each season with pH and organic carbon values
- Total input cost per acre vs. gross income — tracks profitability trends

4.9 Economics of Natural Farming for Cotton

A common concern among farmers considering natural farming is economic viability. The following data from documented studies and farmer cooperatives addresses this concern:

Parameter	Conventional Farming	Natural/Organic Farming (Established)	Transition Period (Year 1-2)
Seed + seed treatment cost	Rs 1,800-2,500/acre	Rs 600-900/acre (non-BT desi + treatment)	Same as natural
Chemical pesticide cost	Rs 2,500-4,500/acre	Rs 0	Rs 500-1,000 (reducing)
Organic input cost	Rs 0	Rs 800-1,500/acre (neem, Trichoderma, etc.)	Rs 1,000-2,000
Fertiliser cost	Rs 3,000-5,000/acre (NPK)	Rs 500-1,000 (bio-fertilisers + FYM)	Rs 1,500-2,500
Total input cost	Rs 7,300-12,000/acre	Rs 1,900-3,400/acre	Rs 3,000-5,500
Average yield	18-22 quintals/acre	15-20 quintals/acre (established)	12-17 quintals/acre
Net income (approx)	Rs 15,000-25,000/acre	Rs 18,000-30,000/acre	Rs 10,000-18,000 (transition loss)

Kisan Tip: The transition period (Year 1-2) is the hardest economically. Farmer groups in Gujarat and Maharashtra who transitioned together shared knowledge, bulk-purchased bio-inputs (reducing cost 30-40%), and accessed organic market premiums sooner through collective certification.

4.10 Farmer FAQ

Q: My plants look healthy — do I still need preventive sprays?

A: Yes. Preventive sprays work before disease establishes. Once symptoms appear, you are already behind — disease has been spreading for 7-14 days invisibly. Think of it like human vaccination: the best time to prevent disease is before it arrives.

Q: How do I know if my soil has enough beneficial microorganisms?

A: Healthy soil smells pleasantly earthy (petrichor), crumbles easily when moist, has earthworms visible when dug, and shows rapid decomposition of added organic matter. For quantitative testing, send soil to an agricultural university for microbial count analysis.

Q: Can I mix organic sprays to save time?

A: Some combinations are fine: neem oil + garlic extract work well together. Others should be avoided: copper oxychloride with any alkaline material. Trichoderma and Pseudomonas can be mixed

in the same drench. When uncertain, apply on alternate days rather than risking inactivation by incompatible mixing.

Q: What is the single most important investment in healthy cotton?

A: Soil organic matter. Every 0.1% increase in soil organic carbon improves water retention, nutrient availability, disease suppression, and root development. Applications of FYM, vermicompost, green manure, and biochar are the highest-return investments in long-term cotton productivity.

Q: Is natural farming viable for medium-scale farmers (10-50 acres)?

A: Yes — and at larger scale the economics improve further. Bulk purchasing of bio-inputs reduces per-acre cost. Farmers at this scale in Gujarat and Maharashtra consistently report 15-25% higher net profit after 3 years of establishment.

Appendix

A.1 Quick Diagnosis & First Action Reference

Symptom Observed	Likely Diagnosis	Immediate Action	First Spray
Angular brown spots on leaves; bacterial ooze	Bacterial Blight	Remove infected leaves; avoid field entry after rain	Copper oxychloride 3 g/L
Upward curl + enations on leaf underside	Curl Virus (CLCuD)	Remove infected plant IMMEDIATELY	Neem oil on surrounding plants
Midday wilt; stem brown inside when cut	Fusarium Wilt	Confirm with stem cut; drench root zone	Trichoderma harzianum 10 g/L drench
Healthy — no symptoms	Healthy Plant	Continue preventive schedule	Neem oil preventive 3 ml/L
Yellow, mottled leaves; stunted growth	Nutrient deficiency or further check needed	Soil test; check roots for galls or rot	Correct deficiency first
Sudden complete wilt; roots fine	Heat or drought stress	Irrigate; shade young plants if possible	No spray needed

A.2 Universal Natural Remedy Kit — 5 Essentials

Every cotton farmer should keep these five items readily available throughout the growing season:

- **Copper Oxychloride (50% WP):** For bacterial blight management and broad antifungal protection — most versatile spray
- **Cold-pressed Neem Oil:** For whitefly control, preventive antifungal protection, and broad-spectrum pest deterrence
- **Trichoderma harzianum WP:** For soil drench to prevent and manage Fusarium wilt and other soil-borne diseases
- **Pseudomonas fluorescens WP:** For soil drench and foliar spray against bacterial and fungal diseases
- **Yellow Sticky Traps:** For whitefly monitoring, population trending, and mass trapping (CLCuD vector)

A.3 Spray Compatibility Chart

Before mixing sprays, check compatibility to avoid inactivation:

Spray A	Spray B	Compatible?	Notes
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Copper Oxychloride	Lime or Wood Ash	NO	Alkaline materials inactivate copper — apply on separate days
Copper Oxychloride	Bordeaux Mixture	NO	Both copper-based — no benefit to combining
Neem Oil	Garlic Extract	YES	Synergistic — good combination
Neem Oil	Pseudomonas fl.	CAUTION	Neem may reduce Pseudomonas viability — apply separately
Trichoderma	Pseudomonas fl.	YES	Can be combined in same soil drench
Trichoderma	Copper-based	NO	Copper kills Trichoderma — never mix
Bordeaux Mixture	Neem Oil	CAUTION	Can be combined but test on few plants first
Garlic Extract	Turmeric + Lime	YES	Compatible, additive effect
Verticillium lecanii	Neem Oil	CAUTION	Neem may affect fungal viability — apply on alternate days

A.4 Disease Comparison Summary Table

Feature	Bacterial Blight	Curl Virus (CLCuD)	Fusarium Wilt
Causal organism	Xanthomonas citri pv. malvacearum (bacterium)	Begomovirus complex (virus)	Fusarium oxysporum f. sp. vasinfectum (fungus)
Transmission	Rain splash, seed, contact	Bemisia tabaci whitefly only	Soil, roots, infected debris
Primary site	Leaves, bolls, stems	Systemically whole plant	Root and vascular system
Key symptom	Angular brown leaf spots, black arm	Upward curl + enations	Midday wilt + brown xylem
Can infected plant be cured?	Partially — with treatment	NO — never	Partially — slow recovery possible
Soil persistence	6-12 months (debris)	Not soil-borne	10-20 years (chlamydospores)
Key management	Copper oxychloride + sanitation	Whitefly vector control	Trichoderma + lime + rotation
Worst weather	Warm, humid, rainy	Hot, dry (whitefly boom)	Warm soil, pH <6.5

A.5 Useful Contacts & Resources

Organisation	Contact	Service
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ICAR Kisan Call Centre	1800-180-1551 (toll-free, 6 AM-10 PM)	Agricultural advisory in local language
CICR Nagpur	+91-712-2500247 cicr@icar.org.in	Cotton-specific research and advisory
State Dept Agriculture Helpline	State-specific (check local KVK)	Input subsidies, crop insurance, advisory
KVK (Krishi Vigyan Kendra)	Find nearest at kvk.icar.gov.in	Local field demonstrations, input supply
NIPHM Hyderabad	www.niphm.gov.in	Pest and disease identification, IPM training
PM Kisan Portal	pmkisan.gov.in	Farmer registration, support schemes

Kisan Mitra AI Knowledge Base v3.0 Cotton Disease Natural Remedy Reference For RAG-powered AI advisory use All remedies are organic and natural Consult your nearest KVK or ICAR for region-specific guidance ICAR Helpline: 1800-180-1551